

I'll use a single citation source so the counts are comparable, and I'll flag the date because they change continuously.

I'm using Semantic Scholar's citation index rather than mixing Google Scholar, IEEE, and publisher counts; otherwise the numbers are not meaningfully comparable.

I've confirmed a consistent set of counts for the main lineage. A few titles have preprint and journal records, so I'm using the consolidated record where the index provides one.

Citation counts below are **Semantic Scholar counts checked June 24, 2026**, except the Informed-MPPI count, which came from the available indexed paper record. Counts vary by database and by preprint/journal-version merging.

Year	Paper	Main contribution	Citations
2018	Information-Theoretic Model Predictive Control: Theory and Applications to Autonomous Driving — arXiv · IEEE	Canonical IT-MPC / MPPI formulation and aggressive-driving application	358
2018	Robust Sampling Based Model Predictive Control with Sparse Objective Information — RSS PDF · DOI	Tube-MPPI: combines MPPI planning with a tracking controller for robustness	89
2019	Informed	CVAE-guided	27

Year	Paper	Main contribution	Citations
	Information Theoretic Model Predictive Control — PDF · IEEE	proposal distributions for better samples	
2021	Robust Model Predictive Path Integral Control: Analysis and Performance Guarantees — arXiv · IEEE	RMPPI: tracking-aware importance sampling, robustness analysis, AutoRally experiments	80
2022	Trajectory Distribution Control for MPPI Using Covariance Steering — arXiv	CC-MPPI: optimizes/control-shapes the trajectory distribution, not only its mean	79
2022	Smooth Model Predictive Path Integral Control without Smoothing — arXiv · IEEE	SMPPI: samples derivative actions to reduce chatter without post-filtering	65
2023	Model Predictive Optimized Path Integral Strategies — arXiv	MPOPI: adaptive importance sampling over a joint control-sequence distribution	26

Year	Paper	Main contribution	Citations
2024	Biased-MPPI: Informing Sampling-Based MPC by Fusing Ancillary Controllers — arXiv · IEEE	Supports arbitrary proposal distributions and fuses learned/classical controllers into sampling	44
2022	Constrained Stochastic Optimal Control with Learned Importance Sampling: A Path Integral Approach — PMC full text · DOI	Learned importance sampling plus constraints; especially relevant to learned sample selection	24
2024	Spline-Interpolated MPPI with Stein Variational Inference for Reactive Navigation — arXiv · IEEE	Spline action parameterization plus SVGD to preserve multiple useful sample modes	12
2022	Autonomous Navigation of AGVs in Unknown Cluttered Environments : log-MPPI Control	Normal-log-normal sampling for feasibility in clutter; includes real AGV experiments	62

Year	Paper	Main contribution	Citations
	Strategy — arXiv · IEEE		

For your specific question about **learning which paths to sample**, I would prioritize: **Informed IT-MPC** → **MPOPI** → **Biased-MPPI** → **Carius et al.** → **Spline/SVGD MPPI**.